

Co-Founder & Lead Engineer: Computational Plasma Physicist

Company: Cascade Dynamics

Status: Stealth-mode, Pre-revenue Deep-Tech Startup

Location: Lab-based in the Deep Tech Hub at Champaign, IL

Role Tier: Founding Team Member / Co-Founder Status

Compensation: Meaningful Founding Equity Package (with transitioning base salary post funding round)

About the Venture

We are a stealth-mode, pre-revenue deep-tech company engineering a new class of physics hardware utilizing plasmas, charged particle beams and light. Powered by a foundational patent portfolio, our platform is developing and testing a proprietary, high-efficiency plasma architecture intended to reduce reliance on hazardous, high-maintenance, and supply-chain-vulnerable legacy technologies across multiple industrial and scientific applications.

Our founding team brings experience in plasma physics, laser optics and entrepreneurship. We are seeking a computational plasma physicist to help model and test candidate coupling pathways among plasmas, lasers, maintained solid particles, highly charged ion production, and potential plasmonic enhancement of the light-matter interaction. Our initial commercial focus is expanding access to heavy-ion single-event-effects testing. We have completed I-Corps customer discovery and were invited by NSF to submit a Phase I SBIR proposal.

The Role

As a Founding Team Member and Lead Computational Plasma Physicist, you will serve as a critical thought partner to the technical founders and the commercial lead. You will own the multiscale computational program that guides architecture decisions from particle response and material release through host-plasma effects, ECR capture and charge breeding, and extraction interfaces.

This is not a purely academic modeling role. You will work directly at the boundary of theory and real-world hardware, using models to design decisive experiments, define diagnostic requirements, and inform accelerator-acceptance and downstream beamline interfaces as the source matures.

Key Responsibilities

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- Multiscale Modeling: Develop and validate a hierarchy of analytic, reduced-order, kinetic, electromagnetic, and particle-transport models. Apply PIC, MD, fluid, or hybrid methods selectively where their scale and assumptions are justified.
- Optical Response and Charge-State Kinetics: Model material- and size-dependent optical absorption, electron emission, heating, and material release. Separately model ECR capture, confinement, ionization kinetics, and charge-state evolution to identify testable coupling pathways.
- Particle-Plasma Interface Modeling: Simulate the charging, heating, vaporization, fragmentation, ionization, residence, and loss of candidate maintained-particle feedstocks. Define stage-specific efficiency metrics, including released-material rate, ECR capture fraction, selected charge-state current, and input power.
- Model Validation and Diagnostic Design: Work closely with the experimental team to compare predictions against calibrated particle-residence measurements, optical and electrical plasma diagnostics, charge-state analysis, current measurements, and energy-distribution diagnostics.
- Core IP & Funding Architecture: Contribute key technical insights, simulation data, and architecture maps to corporate patent portfolios and competitive deep-tech federal grant applications (e.g., SBIR Phase I/II, DARPA, and NSF solicitations).

Required Qualifications

- Educational Background: Ph.D. Candidate (near completion) or recently awarded Ph.D. in Plasma Physics, Computational Physics, Nuclear Engineering, Computer Science, or a closely related quantitative field.
- Simulation Mastery: Proven experience with advanced computational plasma physics tools, electrodynamic solvers, or multi-phase fluid-kinetic packages, including model verification, experimental validation, uncertainty quantification, and reproducible scientific software.
- Domain Expertise: Deep theoretical and practical understanding of plasma chemistry, laser-matter interactions, dusty plasmas, high-vacuum physics, or advanced ion-source physics (e.g., ECR, laser ion sources, EBIS, or SNICS systems).
- Experimental Rigor: Hands-on experience or collaborative fluency with laboratory diagnostics of radiofrequency (RF), microwave-driven, or laser-produced plasmas.
- Mindset: A high-conviction "Moonshot" mindset tailored for a fast-paced, un-siloed "Skunkworks" laboratory environment. Performance dictates corporate progress here.
- Security Clearance: Preferred ability to obtain a U.S. Government Security Clearance (U.S. Citizenship typically required for downstream defense prime deployments).

Preferred Extras

- Computational familiarity tracking multi-stage foil stripping efficiencies or accelerator beam transport matrices (e.g., Monte Carlo/FLUKA codes).
- Experience with custom script implementation in Python, C++, or GPU-accelerated environments for handling heavy diagnostic datasets.

What We Offer

CASDCADE DYNAMICS

- Direct Equity Ownership: A true co-founding technical stake in a venture tackling a multi-billion-dollar bottleneck with definitive, validating customer pull.
- High Impact Culture: The autonomy to build a technical pipeline from scratch alongside world-class peers, completely free of corporate bureaucracy or traditional academic slow-downs.